

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Sixth Semester B.E. (ISE) (Credit System) Degree Examinations

May - June 2023

20IS603 - COMPILER DESIGN

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing Two full questions from Unit – I & Unit – II each and One full question from Unit – III.

Unit – I

Marks BT* CO* PO*

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|-------|---|----|-----|---|---|
| 1. a) | Construct a transition diagram to recognize the following tokens:
i) relation operators
ii) unsigned numbers
iii) identifiers and reserved words | 6 | L*3 | 1 | 2 |
| b) | With an example explain the flow of data in compiler modules. | 8 | L2 | 1 | 1 |
| c) | With a neat diagram, explain the process of program development. | 6 | L2 | 1 | 1 |
| 2. a) | Show that dangling else statement grammar is ambiguous.
i) construct an equivalent unambiguous grammar.
ii) Also Left factor this grammar and check resultant grammar whether ambiguous or not justify your answer. | 10 | L3 | 2 | 2 |
| b) | Write an algorithm to eliminating Left Recursion. Eliminate left recursion from the following grammar:
$bexpr \rightarrow bexpr \text{ or } bterm \mid bterm$
$bterm \rightarrow bterm \text{ and } bfactor \mid bfactor$
$bfactor \rightarrow \text{not } bfactor \mid \text{true} \mid \text{false}$ | 6 | L3 | 2 | 2 |
| c) | With a suitable diagram, explain the lexical analyzer generator using LEX. | 4 | L2 | 2 | 1 |
| 3. a) | What are the drawback in buffer pair buffering techniques? With an example illustrate how it is addressed in sentinels approach? | 8 | L2 | 1 | 1 |
| b) | Write regular definition for:
i) email address ii) unsigned number iii) fax number | 6 | L1 | 1 | 1 |
| c) | Consider the grammar
$bexpr \rightarrow bexpr \text{ or } bterm \mid bterm$
$bterm \rightarrow bterm \text{ and } bfactor \mid bfactor$
$bfactor \rightarrow \text{not } bfactor \mid \text{true} \mid \text{false} \mid (bexpr)$
i) Construct a parse tree for the sentence: not (true or false)
ii) Is this grammar ambiguous? Why? | 6 | L2 | 2 | 1 |

Unit – II

4. a) Consider the following CFG $G = (N = \{S, A, B, C, D\}, T = \{a, b, c, d\}, P, S)$ where the set of productions P is given below:
- $S \rightarrow A$
 $A \rightarrow BC \mid DBC$
 $B \rightarrow Bb \mid \epsilon$
 $C \rightarrow c \mid \epsilon$
 $D \rightarrow a \mid d$
- i) Is this grammar suitable to be parsed using the recursive descendent parsing method? Justify and modify the grammar if needed.
- ii) Compute the FIRST and FOLLOW set of non-terminal symbols of the grammar resulting from your answer in a)

P.T.O.

- iii) Show the stack contents, the input and the rules used during parsing for the input $w = \text{dbb}$
- iv) Construct the corresponding parsing table using the predictive parsing LL method. 12 L3 3 2
- b) Consider the following grammar-
 $E \rightarrow E + E \mid E * E \mid \text{id}$
 i) Construct Operator Precedence Parser.
 ii) Parse the string $\text{id} + \text{id} * \text{id}$
 iii) Find the Operator Precedence Functions. 8 L3 3 1
5. a) Construct LALR (1) parsing table for the given grammar:
 $S \rightarrow L = R$
 $S \rightarrow R$
 $L \rightarrow * R$
 $L \rightarrow \text{id}$
 $R \rightarrow L$
 Also check whether given grammar is ambiguous or not. 14 L4 4 2
- b) Name the data structures used in shift-reduce parser.
 Consider the following grammar
 $S \rightarrow TL$
 $T \rightarrow \text{int} \mid \text{float}$
 $L \rightarrow L, \text{id} \mid \text{id}$
 Parse the input string $\text{int id}, \text{id}$; using a shift-reduce parser. 6 L2 4 1
6. a) Consider the following grammar:
 $S \rightarrow aAd \mid bBd \mid aBe \mid bAe$
 $A \rightarrow c$
 $B \rightarrow c$
 Construct CLR(1) parsing table and check whether it is a CLR(1) grammar or not. 14 L3 4 2
- b) Write a YACC program to evaluate a given arithmetic expression consisting of '+', '-', '*', '/' 6 L2 4 1
- Unit – III**
7. a) Translate the following expression to quadruple, triple and indirect triple: $a + b * c / e \uparrow f + b * c$
 Also construct DAG for the expression:
 (1) $a = b * c$
 (2) $d = b$
 (3) $e = d * c$
 (4) $b = e$
 (5) $f = b + c$
 (6) $g = f + d$ 10 L3 5 2
- b) Explain the issues in designing code generator. 10 L2 5 1
8. a) Write a SDT for numerical representation for the boolean. 10 L2 5 1
- b) Consider following program segment:
 $\text{prod} = 0;$
 $i = 1;$
 do
 {
 $\text{prod} = \text{prod} + a[i] * b[i];$
 $i = i + 1;$
 } while ($i \leq 10$);
 i) Obtain 3 address code. ii) Obtain basic blocks.
 iii) Obtain the flow graph 10 L3 5 2